Feasibility Study

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1. Executive Summary

The Equipment Borrowing Management System (EBMS) is proposed to transform the current manual, paper-based process of borrowing teaching equipment at Eastern International University into a fully digital system. The existing workflow leads to inefficiencies such as data loss, delays, and lack of accountability.

A comprehensive feasibility evaluation demonstrates the project's viability within a five-month development timeline and an estimated total cost of VND 165,000,000. The project aligns with EIU's digital transformation objectives by improving efficiency, accuracy, and transparency.

2. Description of Products and Services

The EBMS is a web-based application designed to manage the complete lifecycle of equipment borrowing—from request submission and approval to issuance and return. It replaces paper forms with an automated process that provides real-time visibility and accountability.

- User Management with role-based access (Admin, IT Staff, Borrower).
- Online Borrowing Requests available anytime.
- Automated Approval Workflow tied to equipment availability.
- Inventory Dashboard displaying real-time equipment status.
- Email Notifications for borrowing confirmations, reminders of upcoming due dates, and verification upon return completion to ensure accountability.
- Reporting and Analytics for usage and performance tracking.
- Audit Log for full traceability of system activity.

3. Technology Considerations

The system will utilize EIU's existing IT infrastructure to minimize setup costs and maximize maintainability. The proposed technology stack includes ASP.NET Core MVC for backend development, Microsoft SQL Server for data storage, and a responsive frontend using HTML5, CSS3, and JavaScript. The system will be hosted on EIU's private cloud or Microsoft Azure and integrated with institutional authentication systems. Security features will include HTTPS encryption, salted password hashing, and role-based access control.

4. Product/Service Marketplace

While the EBMS is designed for internal use, it addresses a growing need in higher education for digitalized resource management. Universities worldwide are increasingly shifting toward centralized IT asset management systems to improve transparency and reduce loss. At EIU, the EBMS will serve as a benchmark for process automation and can later be expanded for campus-wide asset tracking.

5. Marketing Strategy

The EBMS will be promoted internally through awareness campaigns and training workshops. IT staff will conduct demonstrations and share success stories from pilot phases to encourage adoption. Early adopters within departments will act as 'System Champions' to support peers. Clear communication of the system's benefits—such as reduced paperwork, faster approvals, and improved accountability—will foster widespread user engagement.

6. Organization and Staffing

The project will be managed under the supervision of the IT Department. An agile development approach will be used to deliver the Minimum Viable Product (MVP) within four months. The project team will consist of five members with defined roles and responsibilities.

- Project Sponsor – Oversees project funding and progress (Head of IT).
- Project Manager – Leads project planning and coordination (IT Lecturer).
- Developers – Build and test system modules (2 Student Interns).
- QA Engineer – Conducts quality assurance testing (IT Staff).
- IT Support – Handles deployment and maintenance (Technician).

7. Schedule

The estimated implementation period is four months, beginning in November 2025 and concluding in February 2026. Key project phases include initiation, analysis and design, development, pilot testing, and full deployment.

- The project is scheduled for five months (Dec 2025 – Apr 2026) with the following milestones:
- Analysis & Design – 4 weeks (requirements gathering, architecture).
- Development – 8 weeks (coding and internal testing).
- Pilot Testing – 3 weeks (feedback collection and refinement).
- Full Deployment – 2 weeks (campus-wide rollout).

8. Financial Projections

A comprehensive feasibility evaluation demonstrates the project's viability within a five-month development timeline and an estimated total cost of VND 165,000,000. The project aligns with EIU's digital transformation objectives by improving efficiency, accuracy, and transparency.

9. Findings and Recommendations

The feasibility study confirms that the EBMS project is technically feasible, operationally practical, and financially sound. It directly supports EIU's digital transformation strategy and enhances efficiency across academic and administrative operations. Given its high impact and manageable risk profile, it is recommended that EIU proceed with development and pilot implementation, followed by institution-wide deployment after successful evaluation.

Phase	Duration	Key Activities
Initiation	2 weeks	Project setup, approvals, budgeting
Analysis & Design	5 weeks	Requirement gathering and system design
Development	10 weeks	Coding, internal testing, and debugging
Pilot Testing	4 weeks	Trial run, collecting feedback, resolving issues
Training & Refinement	2 weeks	Conduct training sessions and update documentation
Full Deployment	1 week	Launch system across departments

Initial Investment Breakdown

Category	Amount (VND)	Basis of Estimation
Software Development	130,000,000	Estimated based on 2 student developers × 10M/month × 6.5 months including QA time.
Hosting & Cloud Services	8,000,000	Based on EIU Azure Private Cloud annual hosting.
Testing & Training	12,000,000	Covers user testing sessions, training workshops, and materials.
Licensing & Contingency	15,000,000	For potential system licenses and unexpected

		costs.
Total Initial Cost	165,000,000	Sum of all above categories.

Each figure is derived from standard institutional cost estimates, with labor calculated at student internship rates and technical resources priced per EIU's IT infrastructure budget.

Annual Operating Costs and Benefits

Category	Amount (VND)	Basis of Estimation
Maintenance	18,000,000	2 IT staff allocating 5 hours/month × 12 months × 750,000/hour.
Hosting & Email Services	7,000,000	Includes Azure hosting and institutional mail integration.
Total Annual Cost	25,000,000	Sum of annual recurring expenses.
Benefit	Value (VND/year)	Explanation
Labor Savings	70,000,000	Reduced manual workload (~180 hours/month) through automation.
Reduced Equipment Loss	45,000,000	Improved tracking and email alerts reduce missing items.
Total Annual Benefit	115,000,000	Combined operational and financial improvements.

Based on the above data, the estimated Return on Investment (ROI) is 55.8% with a payback period of approximately 1.7 years. The calculations are derived as follows: ROI = (Annual Benefit – Annual Cost) / Initial Investment × 100.